

Mini-CLIP: a laptop-scale reproduction of CLIP-style contrastive image-text learning

1. Paper summary

The reproduced paper is **Learning Transferable Visual Models From Natural Language Supervision** by Radford et al. (2021). The central claim is that visual representations can be learned from natural language supervision at web scale. Instead of training a classifier on a fixed label set such as ImageNet classes, CLIP trains an image encoder and a text encoder to identify which text caption belongs to which image. After pre-training, classification can be expressed as a retrieval problem: class names or prompt templates are encoded as text, images are encoded by the vision tower, and the class with the highest image-text similarity is selected.

This matters because it changes the interface between visual recognition and supervision. A conventional supervised classifier is tied to a predefined taxonomy. CLIP shows that a model trained on broad image-text pairs can transfer to many downstream datasets without dataset-specific supervised training. The paper reports non-trivial zero-shot transfer across many vision benchmarks and shows that natural-language prompts can act as flexible task descriptions.

The novelty is not a new convolution or transformer block. The key contribution is the combination of a simple contrastive objective, a dual-encoder architecture, large-scale noisy image-text data, and prompt-based zero-shot evaluation. In this project we reproduce the mechanism at a smaller scale. The original paper trains on roughly 400 million image-text pairs, while this project trains from scratch on Flickr8k. Therefore the goal is not to match CLIP's absolute numbers. The goal is to reproduce the experimental pattern: learn aligned image and text embeddings, evaluate retrieval, and test prompt sensitivity.

Primary reference:

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning Transferable Visual Models From Natural Language Supervision. ICML 2021.

2. Reproduced result

The closest result in the original paper is Table 13, where Radford et al. report zero-shot text and image retrieval on Flickr30k and MSCOCO. Our project mirrors the structure of that table, but the dataset is changed to Flickr8k and the model is trained from scratch. This makes the comparison methodologically useful but not numerically equivalent.

Caption for the reproduced table:

Reproduction-style adaptation of Table 13 in Radford et al. (2021), evaluated on Flickr8k instead of Flickr30k/MSCOCO.

System	Training data	Eval data	Text R@1	Text R@5	Text R@10	Image R@1	Image R@5	Image R@10
CLIP zero-shot, paper Table 13	Web-scale WIT, about 400M pairs	Flickr30k	88.0	98.7	99.4	68.7	90.6	95.2

System	Training data	Eval data	Text R@1	Text R@5	Text R@10	Image R@1	Image R@5	Image R@10
Mini-CLIP, this project	Flickr8k train split	Flickr8k test	4.10	14.90	20.50	3.84	14.02	21.52

The gap is expected. CLIP has web-scale data, larger encoders, better tokenization, more compute, and evaluates zero-shot transfer from broad pre-training. Mini-CLIP is trained only on Flickr8k, uses a ResNet-18 image encoder and a small Transformer text encoder, and has no ImageNet or CLIP-pretrained weights.

3. Method

3.1 Dual encoder formulation

The model receives a minibatch of matched image-text pairs:

```
(image_1, text_1), ..., (image_N, text_N)
```

The image encoder maps each image to a vector:

```
v_i = f_image(image_i)
```

The text encoder maps each caption to a vector:

```
t_i = f_text(text_i)
```

Both vectors are projected into the same embedding dimension and normalized:

```
z_i = v_i / ||v_i||
u_i = t_i / ||t_i||
```

The similarity between image i and text j is the scaled dot product:

```
s_ij = exp(logit_scale) * z_i dot u_j
```

The learned temperature parameter controls how sharp the similarity distribution is. The implementation clamps the exponentiated scale to a maximum of 100 for numerical stability.

3.2 Symmetric contrastive loss

For a batch of N correct pairs, the diagonal entries of the similarity matrix are positives. All off-diagonal entries are negatives inside the same batch. The loss is computed in both directions.

Image-to-text loss:

```
L_i2t = CE(S, [0, 1, ..., N-1])
```

Text-to-image loss:

```
L_t2i = CE(S^T, [0, 1, ..., N-1])
```

Final loss:

```
L = 0.5 * (L_i2t + L_t2i)
```

This is the same conceptual objective used by CLIP: the model is trained to assign the highest similarity to the matching pair in the batch.

3.3 Model architecture

The image tower is a ResNet-18 from torchvision with random initialization. The classification head is replaced by a linear projection into the shared embedding space.

The text tower is a small Transformer encoder. Captions are lowercased and tokenized with a simple regex tokenizer. The sequence includes <bos> and <eos> markers, and the representation at <eos> is

projected into the shared embedding space.

The final model differs from the original CLIP in several ways:

Component	Original CLIP	This project
Data scale	about 400M image-text pairs	Flickr8k
Image tower	ResNet or ViT family	ResNet-18
Text tower	CLIP Transformer with BPE tokenizer	2-layer Transformer with regex vocabulary
Initialization	trained at web scale	trained from scratch
Main eval	many zero-shot datasets	Flickr8k retrieval and CIFAR-10 prompt ablation

3.4 Course-book connections

The implementation uses building blocks introduced in the course book **Classical Foundations of Artificial Neural Networks**:

- Chapters 21-25 introduce convolutional neural networks, which are the basis for the ResNet-18 image encoder.
- Chapter 26 introduces cross-entropy and maximum-likelihood training, which are used inside the symmetric CLIP loss.
- Chapter 27 introduces modern optimization methods, including the Adam family; the implementation uses AdamW with weight decay.
- Chapter 30 covers PyTorch data loaders, training loops and evaluation, matching the structure of `train.py`, `eval_retrieval.py` and `eval_zeroshot.py`.
- Chapters 37-40 introduce attention, self-attention and Transformers, which are the basis for the text encoder.
- Chapter 42 introduces tokenizers, which connects directly to the simplified caption vocabulary used here.

The project also uses vector representations, dot-product similarity and normalized embeddings throughout the retrieval pipeline. Those ideas connect the earlier linear-model material to modern contrastive representation learning.

4. Experimental setup

4.1 Dataset and splits

The project uses jxie/flickr8k through Hugging Face datasets. Each image has up to five captions. The code uses the train, validation and test splits exposed by the dataset. If the dataset exposes dev instead of validation, the code maps dev to validation.

During training, each image contributes one caption per epoch. The selected caption is rotated deterministically as a function of epoch and example index. This uses more of the caption set over training while keeping the data pipeline simple.

4.2 Preprocessing

Image preprocessing:

- convert to RGB,
- resize and center crop for evaluation,
- random resized crop and horizontal flip for the strong training configuration,
- normalize with CLIP image mean and standard deviation.

Text preprocessing:

- lowercase captions,
- tokenize alphanumeric spans with [a-z0-9]+,
- build vocabulary only from train captions,
- add <bos>, <eos>, <pad> and <unk>,
- pad or truncate to 32 tokens.

4.3 Hyperparameters

The headline run uses configs/flickr8k_strong.yaml.

Hyperparameter	Value
Seed	42
Image size	160
Embedding dimension	256
Text width	256
Text layers	2
Text attention heads	4
Text FF dimension	512
Dropout	0.1
Batch size	128
Epochs	50
Optimizer	AdamW
Learning rate	0.0004

Hyperparameter	Value
Weight decay	0.01
Gradient clipping	1.0
Warmup steps	250
LR schedule	warmup + cosine decay

4.4 Training budget and hardware

The README result was produced on Apple MPS in 2142.9 seconds. The cluster replication committed in results/replication-cluster-001/ records:

- device: CPU,
- wall time: 2983.96 seconds,
- Python: 3.13.9,
- PyTorch: 2.12.1+cu130,
- torchvision: 0.27.1+cu130,
- datasets: 5.0.0.

The code writes package versions into metrics.json, which is included in the committed results folder.

4.5 What was copied and what changed

Copied from the paper:

- dual image/text encoder design,
- symmetric contrastive objective,
- normalized embeddings and learned logit scale,
- retrieval-style evaluation,
- prompt-based zero-shot evaluation idea.

Changed for course constraints:

- Flickr8k replaces web-scale WIT pre-training,
- Flickr8k retrieval replaces Flickr30k/MSCOCO retrieval,
- ResNet-18 and a small Transformer replace full CLIP encoders,
- regex tokenizer replaces CLIP BPE,
- training is from scratch with no CLIP or ImageNet weights,
- CIFAR-10 prompt ablation is limited to a small evaluation subset by default.

5. Results

5.1 Retrieval

The main result is image-text retrieval on Flickr8k.

Model	Eval split	Text R@1	Text R@5	Text R@10	Image R@1	Image R@5	Image R@10
Mini-CLIP strong, original README run	Flickr8k test	4.10	14.90	20.50	3.84	14.02	21.52
Mini-CLIP strong, cluster replication	Flickr8k test	4.10	13.00	20.10	4.00	13.70	21.64

The result is low compared with CLIP, but it is above a random retrieval baseline and shows that the model has learned a measurable shared embedding. The stronger interpretation is qualitative rather than competitive: the CLIP loss and dual-encoder training loop work even at small scale, but scale is essential for the high zero-shot performance reported by Radford et al.

The cluster replication is very close to the original README run. Text R@1 is identical, image R@1 is slightly higher, and the largest retrieval difference is Text R@5, which drops by 1.90 percentage points. This supports the claim that the retrieval result is reproducible within the expected noise of a small contrastive setup.

5.2 Side-by-side with paper

Experiment	Paper result	Our result	Interpretation
Flickr-style text retrieval R@1	88.0 on Flickr30k	4.10 on Flickr8k; 4.10 replicated	Same retrieval format, not same training scale or eval dataset
Flickr-style image retrieval R@1	68.7 on Flickr30k	3.84 on Flickr8k; 4.00 replicated	Large gap caused by data/model/compute scale
Prompted zero-shot classification	Broad zero-shot transfer across many datasets	16.00% on CIFAR-10 subset; 14.00% replicated	Above random chance, but weak transfer

The paper comparison should not be read as a failed attempt to match CLIP. It is evidence for the scaling argument: the same objective is easy to implement and does learn alignment, but the transfer behavior of CLIP depends heavily on massive and diverse pre-training.

6. Ablation

The project includes a prompt ablation inspired by CLIP's prompt-engineering discussion. The trained model is evaluated on CIFAR-10 without supervised fine-tuning. Each class is represented by text prompts, and images are classified by highest image-text similarity.

Caption for the ablation figure/table:

Prompt ablation inspired by the prompt-engineering discussion in Radford et al. (2021), adapted to a Mini-CLIP model trained on Flickr8k.

Prompt variant	Accuracy
Class name only	13.70
a photo of a {label}	16.00
Prompt ensemble	14.20

Cluster replication:

Prompt variant	Accuracy
Class name only	11.40
a photo of a {label}	14.00
Prompt ensemble	12.70

The best prompt in both runs is the simple photo prompt. The ensemble does not improve the result, unlike what is often expected for larger CLIP models. This is a useful negative result: prompt ensembling only helps if the text encoder has learned robust enough semantics for the additional templates to average meaningful class representations.

The replication is consistently lower by 1.5-2.3 percentage points, but the ordering of prompt variants is unchanged. This strengthens the ablation conclusion: prompt wording matters, but the small Flickr8k-trained text encoder does not benefit from prompt ensembling.

7. Limitations and reproducibility notes

Important limitations:

- Flickr8k is tiny relative to CLIP's web-scale pre-training.
- The model is trained from scratch.
- The tokenizer is simple and likely loses useful linguistic structure.
- The image encoder is small.
- The text encoder is small.
- The evaluation datasets do not match the paper exactly.
- Cluster results may differ from Apple MPS results due to backend differences.

Implementation details invented or simplified:

- exact small-model architecture,
- caption rotation rule across epochs,
- augmentation profile,
- tokenizer design,
- CIFAR-10 prompt set,
- output artifact layout.

Sensitivity concerns:

- contrastive learning depends strongly on batch size because in-batch negatives define the classification problem,
- prompt ablation is sensitive to the tokenizer and training captions,
- small-data training can vary across seeds,
- retrieval R@1 is noisy when the dataset and model are small.
- hardware and backend differences matter: the original README result used Apple MPS, while the cluster replication records CPU execution with PyTorch 2.12.1+cu130.

The code fixes the default seed at 42 and writes run metadata, configuration, vocabulary, checkpoints, metrics and generated tables. For a final submission, the exact cluster run directory should be archived or included in the repository release.

The committed cluster replication also provides a small "wow effect" result: retrieval is stable across environments, while zero-shot prompt accuracy drops more noticeably. That contrast suggests the retrieval benchmark is more robust for this small model than the downstream zero-shot transfer test.

8. Reproduction command

The final result is reproduced with:

```
uv run python -m miniclip_repro.reproduce --config configs/flickr8k_strong.yaml --output-dir outputs/flickr8k-strong-
```

Without uv:

```
python -m miniclip_repro.reproduce --config configs/flickr8k_strong.yaml --output-dir outputs/flickr8k-strong-160-b128-
```

Expected artifacts:

- outputs/flickr8k-strong-160-b128/metrics.json,
- outputs/flickr8k-strong-160-b128/checkpoint_best.pt,
- outputs/flickr8k-strong-160-b128/checkpoint_last.pt,
- outputs/flickr8k-strong-160-b128/retrieval_last/retrieval_table.md,
- outputs/flickr8k-strong-160-b128/retrieval_last/retrieval_metrics.json,
- outputs/flickr8k-strong-160-b128/zeroshot_best/prompt_ablation.csv,
- outputs/flickr8k-strong-160-b128/zeroshot_best/prompt_ablation.png.

9. Conclusion

This project reproduces CLIP's core learning mechanism under course-scale constraints. The model learns an aligned embedding space that supports image-text retrieval, and the prompt ablation shows limited but measurable zero-shot behavior. The large gap to the original paper is itself the main scientific lesson: CLIP's method is simple, but its strongest claims emerge from scale, data diversity, and careful evaluation design.

Appendix A. Replication delta summary

The cluster replication gives an additional check beyond a single successful run. The table below compares the original README result with the committed cluster result and reports the absolute difference in percentage points.

Retrieval metric	Original README	Cluster replication	Delta
Text R@1	4.10	4.10	0.00
Text R@5	14.90	13.00	-1.90
Text R@10	20.50	20.10	-0.40
Image R@1	3.84	4.00	+0.16
Image R@5	14.02	13.70	-0.32
Image R@10	21.52	21.64	+0.12

Prompt metric	Original README	Cluster replication	Delta
Class name only	13.70	11.40	-2.30
a photo of a {label}	16.00	14.00	-2.00
Prompt ensemble	14.20	12.70	-1.50

The retrieval deltas are small and mixed in sign, which supports the reproducibility of the retrieval result. The prompt-ablation deltas are consistently negative, but the ranking of prompt variants is unchanged. This distinction is useful scientifically: the embedding alignment measured by retrieval appears more stable than the downstream prompt-based zero-shot transfer test.

Appendix B. Submission checklist

The repository now contains the following submission artifacts:

- README.md describes the paper, target result and one-command reproduction.
- src/miniclip_repro/ contains modular training and evaluation code.
- configs/flickr8k_strong.yaml fixes the main hyperparameters and seed.
- docs/report.md and docs/report.pdf contain the scientific report.
- slides/slides.md and slides/miniclip_reproduction.pptx contain the presentation.
- results/replication-cluster-001/ contains lightweight cluster replication artifacts.

The main remaining manual step before presenting is to rehearse the 10-minute talk and decide which result table to show live. The recommended live demo is not to retrain the model, but to show the one-command reproduction entry point and the committed result artifacts, because the full training run takes close to an hour on the recorded CPU setup.